MTH213 | MAT213 – Elementary Differential Equations

Week 7: Second-Order Differential Equations

Dr Olaoluwa Agboola

Department of Mathematics, Covenant University

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Quick Recap: What We Have Learned So Far

- Definition and classification of differential equations (by type, order, degree and linearity).
- Derivation of differential equations from physical and geometrical problems.
- Techniques for solving **first-order** linear and non-linear equations.
- Applications of first-order equations in growth and decay, cooling, mixing and motion.

Now, we move a step higher — to equations involving the **second derivative**.

Motivation for Studying Second-Order Differential Equations

Why Second Order?

Many real-world systems depend on acceleration (the second derivative of position), such as:

- The motion of a spring–mass system or pendulum ($y'' + ky = 0$).
- The oscillation of electrical RLC circuits.
- Heat conduction and mechanical vibrations.
- Population dynamics involving rates of change of change.

Hence, second-order differential equations are fundamental in modelling physical systems.

Learning Objectives

By the end of this lecture, students should be able to:

- Define and identify second-order differential equations.
- Distinguish between homogeneous and non-homogeneous equations.
- Solve linear homogeneous equations with constant coefficients.
- Solve non-homogeneous equations using suitable methods.

Definition of a Second-Order Differential Equation

Concept

A **second-order differential equation** is an equation involving the **second derivative** of the dependent variable y with respect to the independent variable x.

Example

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$$

Since the highest derivative is the second derivative y'', this is a **second-order** differential equation.

General Form of a Second-Order Differential Equation

Mathematical Expression

A second-order differential equation is generally written as:

$$F(x, y, y', y'') = 0$$

Meaning of Terms

- y — dependent variable (quantity being studied),
- x — independent variable,
- $y' = \dfrac{dy}{dx}$ — first derivative of y,
- $y'' = \dfrac{d^2y}{dx^2}$ — second derivative of y.

Standard (Explicit) Form

Simplified Expression

In many cases, a second-order differential equation can be written in its **standard (explicit) form** as:

$$\boxed{y'' = f(x, y, y')}$$

Interpretation

This means that the second derivative y'' is expressed as a function of:

$$x, \quad y, \quad \text{and } y'.$$

Understanding the Standard Form

Structure of the Function $f(x, y, y')$

The function $f(x, y, y')$ can contain:

- Algebraic expressions in x (e.g., x^2, e^x, $\sin x$, $\ln x$)
- Dependence on y (e.g., $\sin y$, y^3)
- Dependence on y' (e.g., y', $(y')^2$)

Example

$$y'' = x^2 + y' - \sin y$$

Here, y'' is explicitly given in terms of x, y', and y.

Types of Second-Order Differential Equations

- **Linear** — each of y, y', y'' appears to the first power only and are not multiplied together.
- **Nonlinear** — one or more of y, y', y'' appear raised to powers, multiplied together or inside nonlinear functions (like $\sin y$, $e^{y'}$, etc.).

Examples

$$\text{Linear: } y'' + 3y' + 2y = 0$$

$$\text{Nonlinear: } y'' + (y')^2 = \sin y$$

In-Class Activity: Linearity of Second-Order Differential Equations

Instructions

Examine each of the following second-order differential equations carefully. For each equation, determine whether it is **linear** or **nonlinear** and briefly explain your reason.

Equations to Classify

1. $y'' + 5y' + 6y = 0$
2. $y'' + (y')^2 = 0$
3. $y'' + \sin y = 3x$
4. $y'' + xy' + e^x y = 0$
5. $y'' + y\,y' = \cos x$

Second-Order Differential Equations: Concept (1/4)

Understanding the Concept

Second-order linear differential equations are used in science and engineering to describe systems where:

- **Acceleration** matters,
- **Forces** interact, or
- **Energy transfer** occurs.

They model how the *rate of change of a rate*—for example, acceleration as the change of velocity—affects system behavior over time.

Second-Order Differential Equations: Applications

Typical Applications in Real Systems

- **Mechanical Systems:** Vibrations of a mass–spring–damper system or a loaded beam.
- **Electrical Systems:** Voltage and current in *RLC circuits*.
- **Thermal Systems:** Heat conduction and transient temperature variations.
- **Waves and Vibrations:** Oscillations in strings, membranes or sound waves.

All these systems can be represented by second-order linear differential equations.

Second-Order Differential Equations: Example

Mass–Spring–Damper System

The motion of a mass attached to a spring with damping is described by:

$$m\,y''(t) + c\,y'(t) + k\,y(t) = F(t)$$

where each symbol represents a physical parameter of the system.

Second-Order Differential Equations: Interpretation

Parameter Definitions (SI Units)

- m: mass of the object (kilograms, kg)
- c: damping coefficient — resistance to motion (newton-seconds per metre, Ns/m)
- k: spring stiffness (newtons per metre, N/m)
- $F(t)$: externally applied force (newtons, N)

Physical Meaning of Terms

- $m\,y''(t)$: **Inertial force** due to acceleration.
- $c\,y'(t)$: **Damping force** — energy dissipation.
- $k\,y(t)$: **Restoring force** — returns system to equilibrium.

This ODE governs the dynamic response of many oscillatory systems in engineering and physics.

Linear Second-Order Differential Equations

Concept

A **linear second-order differential equation** is one in which the dependent variable y and its derivatives y' and y'' appear only to the **first power** and are not multiplied together.

Example

$$y'' + 3y' + 2y = 0$$

This equation is linear because y, y', and y'' appear to the first power only.

General Form of a Linear Second-Order Differential Equation

General Linear Form

A linear second-order differential equation can be expressed as:

$$a_2(x)\,y'' + a_1(x)\,y' + a_0(x)\,y = r(x)$$

where:

- $a_2(x), a_1(x), a_0(x)$ — known coefficient functions of x;
- y — dependent variable;
- $r(x)$ — known function (called the **non-homogeneous term** or **forcing function**).

Homogeneous and Non-Homogeneous Equations

What Do These Terms Mean?

A second-order linear differential equation is generally written as:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$$

The term $r(x)$ on the right-hand side determines whether the equation is **homogeneous** or **non-homogeneous**.

Homogeneous Differential Equation

Definition

An equation is called **homogeneous** when the right-hand side is zero:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

In this case, there is **no external or forcing function** acting on the system.

Example

$$y'' + 3y' + 2y = 0$$

All terms involve y and its derivatives only — no extra function of x.

Non-Homogeneous Differential Equation

Definition

An equation is called **non-homogeneous (or inhomogeneous)** when the right-hand side is **not zero**:

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$$

Here, $r(x)$ is called the **forcing term** or **driving function**.

Example

$$y'' + 3y' + 2y = e^{-x}$$

The term e^{-x} makes this equation **non-homogeneous** because $r(x) \neq 0$.

In-Class Activity: Homogeneous or Non-Homogeneous?

Instructions

Determine whether each of the following equations is **homogeneous** or **non-homogeneous**.

1. $y'' + 4y = 0$
2. $y'' + 2y' + 3y = \sin x$
3. $y'' - y' = e^{2x}$
4. $y'' + 5y' + 6y = 0$
5. $y'' + 3y' + 2y = x^2$

Hint

Look at the right-hand side $r(x)$:

$$r(x) = 0 \Rightarrow \text{Homogeneous}, \quad r(x) \neq 0 \Rightarrow \text{Non-Homogeneous}.$$

General Solution of Linear Second-Order ODEs

Recall

We have learned to classify equations as:

$$\text{Homogeneous: } r(x) = 0, \quad \text{Non-Homogeneous: } r(x) \neq 0$$

Next Step

Now, let's learn how to **solve** these equations.
Every linear second-order ODE has a general solution of the form:

$$\boxed{y = y_c + y_p}$$

where:

- y_c: Complementary (or homogeneous) solution
- y_p: Particular (or non-homogeneous) solution

Solving the Homogeneous Equation

Standard Form

A **homogeneous linear second-order differential equation with constant coefficients** is written as:

$$a\,y'' + b\,y' + c\,y = 0$$

where a, b, c are constants ($a \neq 0$).

Goal

To find the **general solution** y that satisfies the equation.

Assumed Solution Form

Step 1: Assume a Trial Solution

We assume a solution of the form:

$$y = e^{rx}$$

where r is a constant to be determined.

Step 2: Compute Derivatives

$$y' = \frac{d}{dx}(e^{rx}) = re^{rx}, \qquad y'' = \frac{d^2}{dx^2}(e^{rx}) = r^2 e^{rx}$$

Forming the Characteristic Equation

Step 3: Substitute into the ODE

Substitute $y = e^{rx}$, y', and y'' into

$$a\,y'' + b\,y' + c\,y = 0$$

to get:

$$ar^2e^{rx} + bre^{rx} + ce^{rx} = 0 \Rightarrow e^{rx}\left(ar^2 + br + c\right) = 0$$

Step 4: Simplify

Since $e^{rx} \neq 0$,

$$\boxed{ar^2 + br + c = 0}$$

This is called the **characteristic (or auxiliary) equation**.

Nature of Roots of the Characteristic Equation

The quadratic equation

$$ar^2 + br + c = 0$$

has roots determined by the discriminant:

$$D = b^2 - 4ac$$

Possible Cases

1. $D > 0$: Two distinct real roots r_1, r_2
2. $D = 0$: Repeated real root $r_1 = r_2$
3. $D < 0$: Complex conjugate roots $r = \alpha \pm i\beta$

General Solution Forms

Case 1: Distinct Real Roots ($D > 0$)

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

Case 2: Repeated Real Roots ($D = 0$)

$$y = (C_1 + C_2 x) e^{rx}$$

Case 3: Complex Roots ($D < 0$)

If $r = \alpha \pm i\beta$, then:

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$$

Worked Example: Distinct Real Roots

Example

Solve:

$$y'' - 3y' + 2y = 0$$

Step 1: Write the characteristic equation:

$$r^2 - 3r + 2 = 0$$

Step 2: Factorize:

$$(r-1)(r-2) = 0 \Rightarrow r_1 = 1,\ r_2 = 2$$

Step 3: Write the general solution:

$$\boxed{y = C_1 e^x + C_2 e^{2x}}$$

Worked Example: Repeated Real Roots

Example 2

Solve:

$$y'' - 4y' + 4y = 0$$

Step 1: Write the characteristic equation:

$$r^2 - 4r + 4 = 0$$

Step 2: Factorize:

$$(r-2)^2 = 0 \quad \Rightarrow \quad r_1 = r_2 = 2$$

Step 3: Write the general solution for repeated roots:

$$\boxed{y = (C_1 + C_2 x)e^{2x}}$$

Interpretation: The presence of x multiplying one exponential term ensures we have two linearly independent solutions.

Worked Example: Complex Conjugate Roots

Example 3

Solve the differential equation:

$$y'' + 4y' + 20y = 0$$

This is a second-order linear homogeneous differential equation with constant coefficients.

Step 1: Form the Characteristic Equation

From the equation:

$$y'' + 4y' + 20y = 0,$$

we form the characteristic equation by replacing:

$$y'' \to r^2, \quad y' \to r, \quad y \to 1.$$

Thus, the characteristic equation is:

$$\boxed{r^2 + 4r + 20 = 0}$$

Step 2: Solve the Characteristic Equation

We solve:

$$r^2 + 4r + 20 = 0$$

using the quadratic formula:

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here, $a = 1,\ b = 4,\ c = 20.$

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(20)}}{2}$$

$$r = \frac{-4 \pm \sqrt{16 - 80}}{2}$$

Step 2 (continued): Simplifying the Roots

Continue simplifying:

$$r = \frac{-4 \pm \sqrt{-64}}{2}$$

$$r = \frac{-4 \pm 8i}{2}$$

$$\boxed{r = -2 \pm 4i}$$

Thus, the complex roots have:

$$\alpha = -2 \quad \text{and} \quad \beta = 4.$$

Step 3: General Solution for Complex Roots

If the characteristic roots are:

$$r = \alpha \pm \beta i,$$

then the general solution to the ODE is:

$$\boxed{y = e^{\alpha x}\left(C_1 \cos(\beta x) + C_2 \sin(\beta x)\right)}$$

For this problem:

$$\alpha = -2, \quad \beta = 4.$$

Step 4: Final Solution

Substitute the values $\alpha = -2$ and $\beta = 4$:

$$\boxed{y = e^{-2x}\left(C_1 \cos 4x + C_2 \sin 4x\right)}$$

This is the complete general solution of:

$$y'' + 4y' + 20y = 0.$$

Step-by-Step Summary

1. Write the second-order ODE in the form $ay'' + by' + cy = 0$.
2. Form the characteristic equation $ar^2 + br + c = 0$.
3. Find the roots r_1 and r_2 either by factorizing the characteristic equation or using the quadratic formula
4. Determine the form of the general solution based on the nature of the roots.:
   - Distinct real $\rightarrow y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
   - Repeated real $\rightarrow y(x) = (C_1 + C_2 x)e^{rx}$
   - Complex $\rightarrow y(x) = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$
5. Apply given initial conditions (if any) to find C_1, C_2.

Summary of Homogeneous Solution Forms

At a Glance

Root Type	**Roots**	**General Solution**
Distinct real	$r_1 \neq r_2$	$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
Repeated real	$r_1 = r_2 = r$	$y = (C_1 + C_2 x)e^{rx}$
Complex	$r = \alpha \pm i\beta$	$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

Tip: Always check root type first before writing solution.

Quick Practice (In-Class Activity)

Work in pairs. Solve the following:

1. $y'' + 2y' - 3y = 0$
2. $y'' - 6y' + 9y = 0$
3. $y'' + 9y = 0$
4. $2y'' + 5y' + 3y = 0$

Hint:

- Form the characteristic equation.
- Find the roots.
- Write the general solution using the correct case.

Quick Practice: Final Answers

1. For $y'' + 2y' - 3y = 0$:

$$\boxed{y = C_1 e^{x} + C_2 e^{-3x}}$$

2. For $y'' - 6y' + 9y = 0$:

$$\boxed{y = (C_1 + C_2 x) e^{3x}}$$

3. For $y'' + 9y = 0$:

$$\boxed{y = C_1 \cos 3x + C_2 \sin 3x}$$

4. For $2y'' + 5y' + 3y = 0$:

$$\boxed{y = C_1 e^{-x} + C_2 e^{-\frac{3}{2}x}}$$

Recap: Takeaway

Key takeaway

Every homogeneous second-order linear ODE with constant coefficients can be reduced to solving a simple quadratic equation!

Initial-Value Problem (IVP)

Definition

An **initial-value problem (IVP)** for a second-order differential equation with constant coefficients involves finding a solution of the differential equation that also satisfies two initial conditions at a point x_0:

$$y(x_0) = y_0, \qquad y'(x_0) = y_1$$

where y_0 and y_1 are given constants. These conditions determine a unique solution to the differential equation.

Boundary-Value Problem (BVP)

Definition

A **boundary-value problem (BVP)** for a second-order differential equation requires finding a solution of the differential equation that satisfies conditions at two *different* points:

$$y(a) = \alpha, \qquad y(b) = \beta$$

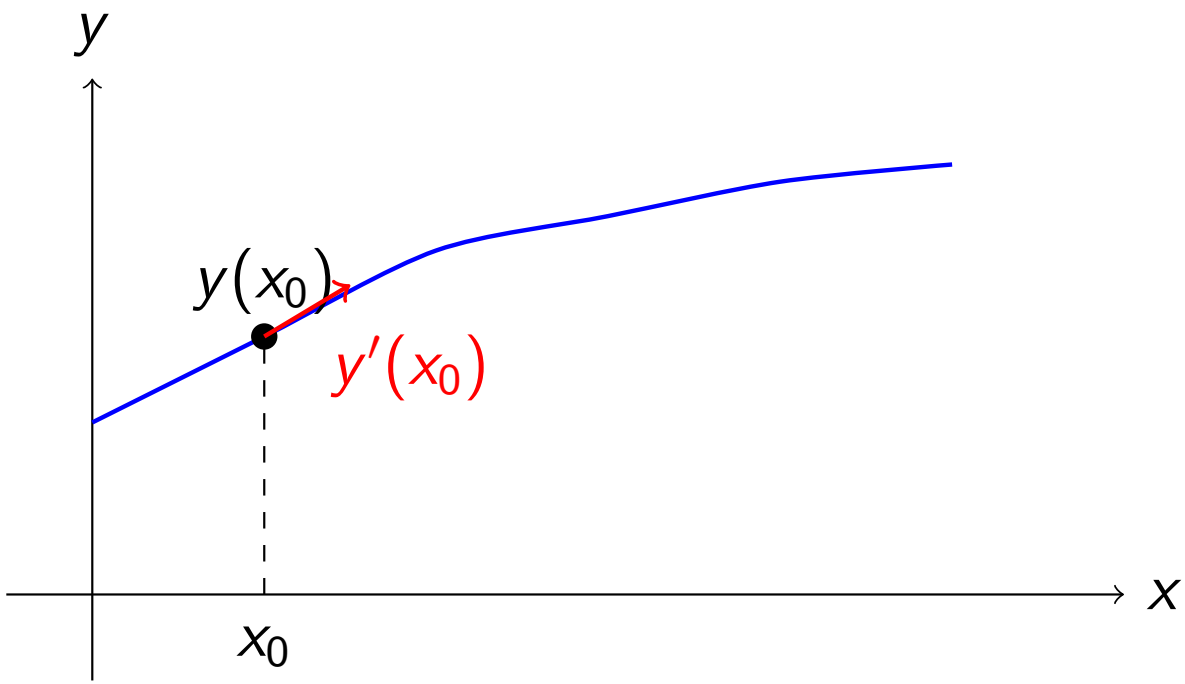
where a and b are points in the interval of interest, and α and β are specified constants. Unlike IVPs, a BVP may or may not have a solution, and when a solution exists, it may not be unique.

IVP vs BVP: Key Differences

Comparison

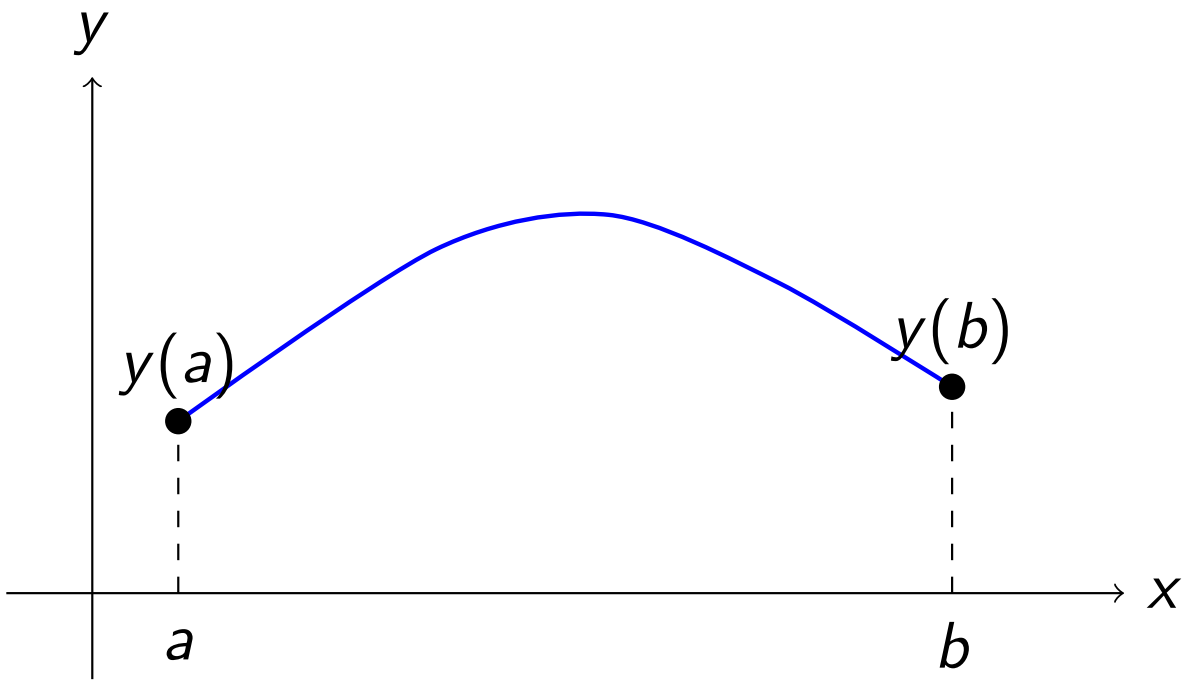
- **Initial-Value Problems (IVPs):**
  - Conditions are given at a *single point*, usually $x = x_0$.
  - Typically have a *unique* solution (for well-posed linear ODEs).
- **Boundary-Value Problems (BVPs):**
  - Conditions are given at *two different points*, usually $x = a$ and $x = b$.
  - A solution may exist or may not exist.
  - Even when a solution exists, it may *not be unique*.

Diagram: Initial-Value Problem (IVP)



IVP: Both $y(x_0)$ and $y'(x_0)$ are specified at the *same point*.

Diagram: Boundary-Value Problem (BVP)



BVP: Conditions $y(a)$ and $y(b)$ are specified at *two different points*.

Example: Initial-Value Problem

Problem

Solve the initial-value problem:

$$y'' + y' - 6y = 0, \quad y(0) = 1, \; y'(0) = 0$$

Step 1: Form the Characteristic Equation

Step 1: Replace y'' with r^2, y' with r, and y with 1 to form the characteristic equation:

$$r^2 + r - 6 = 0$$

Remark: This converts the differential equation into a quadratic equation.

Step 2: Solve the Quadratic Equation

Step 2: Solve for r using the quadratic formula or factorization:

$$r^2 + r - 6 = 0 \implies (r+3)(r-2) = 0$$

$$\Rightarrow r_1 = 2, \quad r_2 = -3$$

Remark: The roots are real and distinct.

Step 3: Write the General Solution

Step 3: For real and distinct roots, the general solution of the homogeneous equation is:

$$y(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

Substitute the roots:

$$y(x) = C_1 e^{2x} + C_2 e^{-3x}$$

Remark: C_1 and C_2 are constants to be determined using the initial conditions.

Step 4: Differentiate to Apply Initial Conditions

Step 4: Differentiate the general solution:

$$y'(x) = 2C_1e^{2x} - 3C_2e^{-3x}$$

Apply the initial conditions: Condition 1: $y(0) = 1$

$$y(0) = C_1e^{2\cdot 0} + C_2e^{-3\cdot 0} = C_1 + C_2$$

$$\implies C_1 + C_2 = 1$$

Condition 2: $y'(0) = 0$

$$y'(x) = 2C_1e^{2x} - 3C_2e^{-3x} \implies y'(0) = 2C_1 - 3C_2$$

$$\implies 2C_1 - 3C_2 = 0$$

Remark: We now have a system of two equations with two unknowns.

Step 5: Solve for the Constants

Step 5: Solve the system:

$$\begin{cases} C_1 + C_2 = 1 \\ 2C_1 - 3C_2 = 0 \end{cases}$$

From the second equation: $2C_1 = 3C_2 \implies C_1 = \frac{3}{2}C_2$
Substitute into the first equation:

$$\frac{3}{2}C_2 + C_2 = \frac{5}{2}C_2 = 1 \implies C_2 = \frac{2}{5}$$

$$C_1 = \frac{3}{2}C_2 = \frac{3}{5}$$

Step 6: Final Solution

Step 6: Substitute C_1 and C_2 back into the general solution:

$$\boxed{y(x) = \frac{3}{5}e^{2x} + \frac{2}{5}e^{-3x}}$$

Example: Boundary-Value Problem

Problem

Solve the boundary-value problem:

$$y'' + 2y' + y = 0, \quad y(0) = 1,\ y(1) = 3$$

Step 1: Form the Characteristic Equation

Step 1: Replace y'' with r^2, y' with r, and y with 1:

$$r^2 + 2r + 1 = 0$$

Remark: This is a quadratic equation derived from the differential equation.

Step 2: Solve the Quadratic Equation

Step 2: Factorize the quadratic:

$$r^2 + 2r + 1 = (r+1)^2 = 0$$

$$\Rightarrow r = -1 \quad \text{(repeated root)}$$

Remark: Since the root is repeated, the solution involves both e^{-x} and xe^{-x}.

Step 3: Write the General Solution

Step 3: For a repeated root $r = -1$, the general solution is:

$$y(x) = (C_1 + C_2 x)e^{-x}$$

Remark: C_1 and C_2 are constants to be determined from the boundary conditions.

Step 4: Apply the Boundary Conditions

Step 4: Apply $y(0) = 1$:

$$y(0) = (C_1 + C_2 \cdot 0)e^0 \implies C_1 = 1$$

Apply $y(1) = 3$**:**

$$y(1) = (C_1 + C_2 \cdot 1)e^{-1} \implies (1 + C_2)e^{-1} = 3$$

Solve for C_2**:**

$$1 + C_2 = 3e \implies C_2 = 3e - 1$$

Step 5: Final Solution

Step 5: Substitute the constants into the general solution:

$$\boxed{y(x) = \big(1 + (3e - 1)x\big)e^{-x}}$$

Remark: This is the solution to the boundary-value problem.

In-Class Activity

Work in pairs and solve the following problems:

1. Solve the initial-value problem:

$$y'' - 3y' - 10y = 0; \quad y(0) = 0,\ y'(0) = 7$$

2. Solve the boundary-value problem:

$$y'' + y = 0; \quad y(0) = 0,\ y\left(\frac{\pi}{4}\right) = 1$$

Hints:

- Form the characteristic equation for each problem.
- Find the roots of the characteristic equation.
- Write the general solution according to the type of roots (real distinct, repeated, or complex).
- Apply the initial or boundary conditions to determine constants.

Problem 1: Step 1

Form the characteristic equation:

$$r^2 - 3r - 10 = 0$$

Solve for roots:

$$r = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-10)}}{2} = \frac{3 \pm \sqrt{9 + 40}}{2} = \frac{3 \pm 7}{2}$$

$$r_1 = 5, \quad r_2 = -2$$

Problem 1: Step 2 - General Solution

The general solution for real distinct roots:

$$y(x) = C_1 e^{5x} + C_2 e^{-2x}$$

Differentiate to apply initial conditions:

$$y'(x) = 5C_1 e^{5x} - 2C_2 e^{-2x}$$

Problem 1: Step 3 - Apply Initial Conditions

Apply $y(0) = 0$:

$$C_1 + C_2 = 0 \implies C_2 = -C_1$$

Apply $y'(0) = 7$:

$$5C_1 - 2C_2 = 7 \implies 5C_1 - 2(-C_1) = 7 \implies 7C_1 = 7$$

$$\implies C_1 = 1, \; C_2 = -1$$

Final solution:

$$\boxed{y(x) = e^{5x} - e^{-2x}}$$

Problem 2: Step 1 - Characteristic Equation

Form the characteristic equation:

$$r^2 + 1 = 0 \implies r = \pm i$$

General solution for complex roots:

$$y(x) = C_1 \cos x + C_2 \sin x$$

Problem 2: Step 2 - Apply Boundary Conditions

Apply $y(0) = 0$:

$$C_1 \cos 0 + C_2 \sin 0 = C_1 = 0 \implies y(x) = C_2 \sin x$$

Apply $y\left(\frac{\pi}{4}\right) = 1$:

$$C_2 \sin \frac{\pi}{4} = 1 \implies C_2 \cdot \frac{\sqrt{2}}{2} = 1 \implies C_2 = \sqrt{2}$$

Final solution:

$$\boxed{y(x) = \sqrt{2} \sin x}$$

Take-Home Assignment: Second-Order ODEs (Part I)

Instructions: Find the general solution for each of the following differential equations. For any given initial conditions, find the particular solution.

1. $y'' = -4y'$
2. $y'' = 4y'$
3. $36y'' - 36y' = -13y$
4. $3y'' + 2y' = 0$
5. $6y'' + 6y' + 13y = 0; \quad y(0) = 2,\ y'(0) = 1$
6. Solve the second-order differential equation:

$$\frac{d^2y}{dx^2} - a^2y = 0$$

and show that the general solution can be written as:

$$y = A\cosh(ax) + B\sinh(ax)$$

Take-Home Assignment: Second-Order ODEs (Part II)

Solve the boundary-value problem:

$$y'' - 4y = 0$$

with boundary conditions:

$$y(0) = 3, \qquad y(2) = 1.$$